

ADITYA CHUNDURI

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SUMMARY

AI engineer and independent researcher who designs, builds, and ships systems end-to-end across LLM agent security, retrieval-augmented generation, computer vision, and multimodal AI. First-authored a Zenodo preprint introducing a transparency-aware evaluation protocol for tool-using agents, implemented as Agent Shield on the Inspect AI harness. Builds with a research mindset — reproducible pipelines, seed-pinned evaluations, full test coverage, and public repos — and is equally comfortable shipping product-facing APIs. M.S. Applied Data Science, USC. U.S. work authorized.

EDUCATION

University of Southern California

M.S., Applied Data Science

Los Angeles, CA

Jan 2024 – Dec 2025

SRM Institute of Science and Technology

B.Tech., Computer Science & Engineering, AI/ML Specialization

Chennai, India

Jun 2019 – Jul 2023

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript/TypeScript, Bash, C/C++, Scala/Spark (academic)

LLM & Agents: LLM agents, RAG, embeddings, context construction, prompt injection testing, evals, Inspect AI, AgentDojo, LangChain, Hugging Face, Ollama

AI Security: Prompt injection, MCP tool poisoning, RAG/memory poisoning, exfiltration, social engineering, behavioral drift, OWASP LLM, OWASP Agentic, MITRE ATLAS, AIVSS

ML / CV / Audio: PyTorch, TensorFlow/Keras, scikit-learn, OpenCV, U-Net, YOLO, Diffusers, librosa, Demucs, Beat This!, CLAP, Essentia, model evaluation, experiment tracking

Engineering: FastAPI, React/TypeScript, Flask, Streamlit, Docker, pytest, GitHub Actions, CI/CD, OpenTelemetry, PostgreSQL, SQLite, Neo4j, ChromaDB

PROJECTS

Agent Shield (Python, Inspect AI, AgentDojo, ChromaDB, pytest, GitHub Actions)

[GitHub](#) | [preprint](#)

- Authored an open-source LLM agent evaluation framework on Inspect AI covering 6 live attack modules and 24 attack IDs across prompt injection, MCP tool poisoning, RAG and memory poisoning, covert exfiltration, social engineering, and multi-turn behavioral drift.
- Introduced Transparency Rate, a new metric for whether an agent flags an attack to its operator, alongside Benign Utility, Utility Under Attack, and Targeted ASR in a six-cell outcome model that extends AgentDojo's standard 2x2 matrix.
- Ran seed-pinned evaluations across 4 models (Claude Sonnet 4.5, Llama 3.3 70B, Llama 3.1 8B, Gemini Flash) with Wilson 95% CIs at $n=20$; three of four score $TR=0.000$ on the anchored injection row; spotlighting raises TR to 0.800 but increases ASR by 0.200 via paraphrase residue.
- Mapped every attack to OWASP LLM, OWASP Agentic, MITRE ATLAS, and AIVSS, with pytest coverage, GitHub Actions CI, and reproducibility artifacts; released as a preprint with framework, attack registries, seeds, and logs.

AI RemixMate (FastAPI, React/TypeScript, Demucs, Beat This!, CLAP, librosa, Essentia, pyloudnorm)

[GitHub](#)

- Built a full-stack DJ engine: FastAPI async job queue with SQLite write-through persistence, React/TypeScript frontend (8 pages, SSE live job streaming, Zustand state), and an automated -14 LUFS mastering pipeline — 374+ tests across unit, behavioral, and integration suites with GitHub Actions CI.
- Implemented a research-grade MIR core: TIV harmonic scoring (Bernardes et al. 2016 Tonal Interval Space), Beat This! (ISMIR 2024) downbeat detection with 8/16/32-bar grid snapping, cosine-taper stem bass ramp with no IIR bleed, and FxNorm per-stem-type LUFS normalization using corpus-derived targets — all replacing earlier librosa/IIR approximations.
- Added CLAP 512-D semantic search (crate_digger.py; 35-D keyword fallback when CLAP absent), Essentia arousal/valence energy arc modeling for setlist planning, and rekordbox XML + Serato GEOB cue export to bridge RemixMate output directly into professional DJ software.

AkashicTree (Ollama, Diffusers, Replicate FLUX.1, ElevenLabs, Coqui TTS, FFmpeg)

[GitHub](#)

- Built a model-agnostic multimodal generation pipeline that converts topic prompts into reviewed scripts, storyboard prompts, generated images, narrated audio, and FFmpeg-assembled vertical videos.
- Supported tiered local and cloud backends using Ollama, Diffusers/Stable Diffusion, Coqui TTS, FLUX.1 via Replicate, and ElevenLabs.

AI Health Journal (Flask, Ollama, ChromaDB)

[GitHub](#)

- Built a fully local Flask/Ollama RAG assistant with ChromaDB retrieval, session memory, and prompt-injection-over-retrieval tests for grounded responses — no data leaves the device.
- Compared dense, BM25, RRF hybrid, and cross-encoder reranked retrieval on a provisional gold set, measuring MiniLM hybrid+rerank nDCG ~0.78 at 504 ms versus hashing hybrid at ~0.8 ms.

MetaLearnML (scikit-learn, PyTorch, Neo4j, FastAPI)

[GitHub](#)

- Designed an AutoML-style model-selection prototype that infers task type, recommends preprocessing strategies, and ranks candidate models from prior experiment history.
- Implemented a Neo4j experiment graph, FastAPI service layer, and automated JSON/Markdown reports for reproducible model-selection comparisons.

Model Behavior Lab (Python, Ollama, pandas, JSON)

[GitHub](#)

- Wrote an Ollama and pandas behavior-scoring harness for prompt-scenario analysis, an early precursor to Agent Shield's transparency evals, structuring model outputs into JSON and CSV artifacts.

Cognitive Signal Benchmark (Python, Kaggle / Google DeepMind AGI hackathon)

[Kaggle](#)

- Published a public benchmark and caught an SDK assertion bug that inflated leaderboard scores, reported with a reproducible case.

WORK EXPERIENCE

USC Viterbi School of Engineering

Los Angeles, CA

Research Assistant, Computer Vision & Medical Imaging

Aug 2024 – Dec 2024

- Engineered a TensorFlow/Keras U-Net experimentation pipeline for retinal artery-vein segmentation on DRIVE and CHASE_DB1 fundus datasets, with per-run AUC logging from ~0.93 to 0.97 across model variants.
- Surveyed published retinal segmentation benchmarks to define AUC baselines; integrated rotation, flip, and contrast augmentation into the preprocessing pipeline, raising validation AUC by ~0.04 over the non-augmented baseline.
- Developed reusable preprocessing, validation, and model-comparison workflows with Keras history and TensorBoard artifacts, enabling reproducible experiments across architecture versions and producing structured per-experiment reports for faculty review.

SSN College of Engineering

Remote

Research Intern, Computer Vision & Object Detection

Jun 2021 – Jul 2021

- Built YOLO-based object-tracking prototypes and reusable PyTorch/TensorFlow modules for detection, preprocessing, tracking, and experiment iteration across video sequences.
- Annotated and curated a 1,200+ frame video dataset with ground-truth bounding boxes; benchmarked YOLOv5 against YOLOv4 on a held-out test set using mAP@0.5, and documented findings in a structured report submitted to the research group.

PUBLICATIONS

Chunduri A., "Beyond Attack Success Rate: Measuring Operator-Facing Transparency in LLM Agent Security," *Zenodo* preprint, Jun. 2026. [doi:10.5281/zenodo.20789431](https://doi.org/10.5281/zenodo.20789431)

Chunduri A. et al., "Wind Power Forecasting Using Machine Learning," *IJRASET*, Aug. 2023.

CERTIFICATIONS & COURSES

IBM Data Science & AI Professional Certificates | Machine Learning — Stanford University (Coursera) | Introduction to Computer Vision & Image Processing — IBM (Coursera) | Algorithmic Toolbox — UC San Diego (Coursera)